Topic: Homogeneous higher order equations

Question: Find the general solution to the third order homogeneous differential equation.

$$y''' - 6y'' + 11y' - 6 = 0$$

Answer choices:

A $y(t) = c_1 t^2 e^t + c_2 e^{6t} + c_3 e^{11t}$

B $y(t) = c_1 e^t + c_2 e^{2t} + c_3 e^{3t}$

C $y(t) = c_1 e^{-6t} + c_2 e^{6t} + c_3 e^{11t}$

D $y(t) = c_1 + c_2 t e^{2t} + c_3 t^2 e^{3t}$

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Solution: B

We'll write the characteristic polynomial associated with the differential equation, then factor it.

$$r^3 - 6r^2 + 11r - 6 = 0$$

$$(r-1)(r-2)(r-3) = 0$$

The equation has distinct real roots $r_1 = 1$, $r_2 = 2$, and $r_3 = 3$. The root $r_1 = 1$ will contribute $c_1 e^t$ to the solution, $r_2 = 2$ will contribute $c_2 e^{2t}$, and $r_3 = 3$ will contribute $c_3 e^{3t}$, so the general solution is the sum of all three,

$$y(t) = c_1 e^t + c_2 e^{2t} + c_3 e^{3t}$$

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Topic: Homogeneous higher order equations

Question: A fifth order homogeneous differential equation has roots $r_1 = -4$, $r_2 = -4$, $r_3 = -4$, and r_4 and r_5 are roots that are not equal to -4. What will the portion of the solution that accounts for r_1, r_2, and r_3 look like?

Answer choices:

A $c_1e^{-4t} + c_2te^{-4t} + c_3e^{5t}$

B $c_1e^{-4t} + c_2e^{-4t} + c_3e^{-4t}$

C $c_1e^{-4t} + c_2t^{-4}e^{t} + c_3t^{4}e^{t}$

D $c_1e^{-4t} + c_2te^{-4t} + c_3t^{2}e^{-4t}$

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Solution: D

When the differential equation has k equal real roots (call these roots r_1), the portion of the general solution that accounts for those roots will be

$$c_1 e^{r_1 t} + c_2 t e^{r_1 t} + \ldots + c_k t^{k-1} e^{r_1 t}$$

In this problem, the root -4 appears three times. In other words, we have three equal roots, $r_1 = r_2 = r_3 = -4$. Plugging $k = 3$ and $r_1 = -4$ into the formula above, the portion of the general solution that accounts for the equal real roots $r_1 = r_2 = r_3 = -4$ will be

$$c_1 e^{-4t} + c_2 t e^{-4t} + c_3 t^2 e^{-4t}$$

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Topic: Homogeneous higher order equations

Question: Given the sixth order homogeneous differential equation and its associated characteristic polynomial, find the general solution to the differential equation.

$$y^{(6)} - 10y^{(5)} + 48y^{(4)} - 132y''' + 221y'' - 210y' + 98y = 0$$

$$(r^2 - 2r + 2)(r^2 - 4r + 7)^2 = 0$$

Answer choices:

A $y(t) = c_1 e^t \cos t + c_2 e^t \sin t$

$$+c_3 e^{2t} \cos(\sqrt{3}t) + c_4 e^{2t} \sin(\sqrt{3}t) + c_5 te^{2t} \cos(\sqrt{3}t) + c_6 te^{2t} \sin(\sqrt{3}t)$$

B $y(t) = c_1 \cos t + c_2 \sin t$

$$+c_3 e^{4t} \cos(\sqrt{7}t) + c_4 e^{4t} \sin(\sqrt{7}t) + c_5 t^2 e^{4t} \cos(\sqrt{7}t) + c_6 t^2 e^{4t} \sin(\sqrt{7}t)$$

C $y(t) = c_1 e^t + c_2 e^{-2t} + c_3 e^{2t} + c_4 te^t + c_5 e^{-4t} + c_6 e^{7t}$

D $y(t) = c_1 e^t \cos t + c_2 e^t \sin t$

$$+c_3 e^{3t} \cos(2t) + c_4 e^{3t} \sin(2t) + c_5 te^{3t} \cos(2t) + c_6 te^{3t} \sin(2t)$$

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Solution: A

Solving the characteristic polynomial gives

$$r^2 - 2r + 2 = 0$$

$$r^2 - 2r + 1 - 1 + 2 = 0$$

$$(r-1)^2 = -1$$

$$r = 1 \pm i$$

and

$$r^2 - 4r + 7 = 0$$

$$r^2 - 4r + 4 - 4 + 7 = 0$$

$$(r-2)^2 = -3$$

$$r = 2 \pm \sqrt{3}i$$

The roots $r = 1 \pm i$ have multiplicity 1, and the roots $r = 2 \pm \sqrt{3}i$ have multiplicity 2. Which means the roots $r_{1,2} = 1 \pm i$ will contribute

$$c_1 e^{\alpha t} \cos(\beta t) + c_2 e^{\alpha t} \sin(\beta t)$$

$$c_1 e^{t} \cos t + c_2 e^{t} \sin t$$

to the general solution, while the roots $r_{3,4} = 2 \pm \sqrt{3}i$ and $r_{5,6} = 2 \pm \sqrt{3}i$ will contribute

$$c_3 e^{\alpha t} \cos(\beta t) + c_4 e^{\alpha t} \sin(\beta t) + c_5 t e^{\alpha t} \cos(\beta t) + c_6 t e^{\alpha t} \sin(\beta t)$$

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$$c_3e^{2t}\cos(\sqrt{3}t) + c_4e^{2t}\sin(\sqrt{3}t) + c_5te^{2t}\cos(\sqrt{3}t) + c_6te^{2t}\sin(\sqrt{3}t)$$

Therefore, the general solution to the sixth order homogeneous differential equation is

$$y(t) = c_1e^t\cos t + c_2e^t\sin t$$

$$+c_3e^{2t}\cos(\sqrt{3}t) + c_4e^{2t}\sin(\sqrt{3}t) + c_5te^{2t}\cos(\sqrt{3}t) + c_6te^{2t}\sin(\sqrt{3}t)$$

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